

Working With Clue Words and K4 Shifts

k4shiftsVSresequencing.pdf

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(See last page for K4 shifts)

Before going any further, note that KRYPTOS = 1473625, and 32514 is the same without the 7 and 6, but in the same order, otherwise. This is important as we need to hold to this regime.

32514 DYHR - 12345 HYDRA
32514 DIAWI - 12345 WIDIA
32514 INFBN - 12345 BNINF

But remember:

1234512345 WIDIAWIDIA has DIAWI (pos. 3-7) now shows a reversal of values of the I and I. Same with INFBN, but with the N and N, instead.

Working with DIAWI for now...

DIAWA appears initially in K4 shift 0 in pos. 21-25. DIAWA appears again at K4 shift 21, pos. 25-29, Assigning DIAWI values 14325 (instead of 12345 because of the reversals of the values of the I and I) gives:

14325 DIAWI - 12345 DWAI

DWAI appears in pos. 26-30 in the 2nd row of K4, shift 16 (0-16).

Now dealing with INFBN.

The same thing happens with INFBN whose initial appearance is at K4 shift 5, 3rd row (pos. 21-25). It appears again at K4 shift 0 in pos. 25-29 of third row. IBFNN appears in K4 shift 21 pos. 26-30, 2nd row.

NOW...

DIAWI appears initially in shift 0 of K4. With a K4 shift of 5, the result is INFBN in the same position. INFBN does appear with DIAWI in shift 0, but that does not appear to be the origin of INFBN, but the result of a shift of -5 to K4 shift 0, pos. 25-30. THEN as shift to K4 shift 21 occurs, resulting in the appearance of IBFNN.

SO...

The scheme appears to be that, without looking further at K4, two strings, DIAWI and INFBN end up being converted to DWAI and IBFNN, both of which end in doublets following this rule:

- a. The original string appears in K4 in its original K4 shift (0-26), pos. 21-25. (DIAWI at shift 0, INFBN at shift 5)
- b. The same string is then shifted to K4 shift (original shift -5). (DIAWI $0(26) - 5 = 21$, INFBN $5 - 5 = 0$)
- c. A converted form of the same characters, but ending in a doublet of the two common characters appears shifted again -5. (DWAI $21 - 5 = 16$, IBFNN $0(26) - 5 = 21$)

In other words, the original string appears in positions 21-25 of the third row of a K4 shift (0 through 25). It is then shifted a -5 K4 shift and appears in positions 25-29 of the third row. Finally, it appears after another -5 K4 shift in positions 26-30 of the 2nd row after having undergone a transposition that reverses the values of the two similar characters (places the second character in the fourth position and places the fourth character in the second position according to the rule that appears in the third paragraph on page 1.

Now, it can be shown that this scheme occurs for all shifts of K4. IE, the string residing in the 3rd row, pos. 21-25, OXOOX, will, after two shifts and a transposition of the values in the Xes (2 and 4), will end up as OOOXX in Row 2, pos. 26-30. There will be two K4 shifts of -5 each.

One has to ask, will the reverse occur if I take the first doublet and preceding 3 characters of K4 shift 0, LIFBB, and attempt to arrive at LBFIB by shifting +5 twice? The answer is NO. In fact this procedure does not work for any doublet-ended strings in K4 shift 0 except for the NGKSS string in the 2nd row, pos. 26-30. Reversing back from NGKSS, we arrive initially at K4 shift 10, pos. 21-25 (as we expected).

How about K4 shift 16 where DWAI lies? Will BHLGG produce BGLHG if we reverse the process? The answer is again NO. Nor will any 5-character strings ending in doublets in locations other than the 2nd row, pos. 26-30.

There is one odd occurrence where two strings exhibiting the structure OXOOX are together and offset from the other by one position. They are in K4 shift 0 starting at 3rd row, position 3. They are TQSJQ and QSJQS. These sort of stood out as I was scanning everything looking for any cases where the OXOOX situation existed. I could find no relationship between them and the other connections that I made.

So, The bottom line at this point seems to be that the characters surrounding the CT implied to look for a way to complete Hoover's name. A shift of 21 was applied that resulted in the OOH appearing. The positive result of this actually could have caused me to look at the adjacent letters around OOH. Together, they are QOOHUAKFR. However, I did a 21 shift of all of K4 and found DIAWI in another place. This made me want to see if other shifts got results. They did. A shift of 5 found INFBN, which I saw merged with DIAWI in the 0 shift of K4. This confused me at first, as I was thinking that DIAWI and INFBN were related in a way other than the Caesar shift of 5. IE a shift of 21 is encode and a shift of 5 is decode since $26 - 21 = 5$. I now think INFBN was chosen as a clue to use shifts of 5. This caused me to arrive at a shift of 16 which had DWAI in it. I looked at the values and rationalized how DWAI could come from DIAWI and then remembered the switching of the values of the I and I according to what I discovered about lining up WIDIAWIDIA/1234512345, getting DIAWI now worth 14325 instead of 12345. Now it made some sense.

Now, we have for all shifts of K4, An origin for a string in the form of OXOOX lying in pos. 21-25 of the 3rd row of K4 shift Y, then a Y-5 K4 shift with OXOOX lying in pos. 25-29 of row 3, then a final -5 K4 shift (to Y-10) but with a transposition of characters to the form OOOXX in pos. 26-30 of the 2nd row. The regime is valid for all 26 K4 shifts. But this scheme is valid only in the positions I mentioned above:

Y - 0 OXOOX positions 21-25 in Row 3
y - 5 OXOOX positions 25-29 in Row 3 (a K4 shift of -5)
y - 10 OOOXX positions 26-30 in Row 2 (a K4 shift of -5)

I don't feel that DIAWI or INFBN are specifically pertinent. I think he used them to map out the shifts. Interesting that he identified them as relating to Hoover, which confirmed that solving for OOH on the outset was essentially a necessary step in moving forward.

What to consider as the next step? After manipulating with two shifts and a transposition resulting in strings ending in doublets, I would have thought that keeping the doublets out of the resulting substitution routine would be prudent. But we don't know what K4 CT says except for NYPVTTMZFPK says BERLINCLOCK. And the TT corresponds with the SS in ABSCISS (since a Z has replaced the ? at the front of K4 CT).

Three things are going on that show up involving shifts of 5.

1. Out of all 26 shifts of K4 (0 through 25), five positive shifts have been chosen as original positions:
 - a. The original position of DIAWI is at shift $\boxed{0}$.
 - b. The original position of INFBN is at shift $0 + 5 = \boxed{5}$.
 - c. The original position of NSKGS is at shift $5 + 5 = \boxed{10}$.
2. Out of all 26 shifts of K4 (0 through 25), SHIFTS OF -5 from original positions have been chosen as intermediate positions (numbers below are shifts):
 - a. The intermediate position of DIAWI is at a shift of -5 from the original position $0 (=26)$, $26 - 5 = \boxed{21}$.
 - b. The intermediate position of INFBN is at shift of -5 from the original position 5, $5 - 5 = \boxed{0}$.
 - c. The intermediate position of NSKGS is at a shift of -5 from the original position 10, $10 - 5 = \boxed{5}$.
3. Out of all 26 shifts of K4 (0 through 25), SHIFTS OF -5 from intermediate positions have been chosen as destination positions and their strings have undergone transpositions from 12345 to 14325.
 - a. The destination position of DIAWI and transposition to DWAI is at a shift of -5 from the intermediate position of 21, $21 - 5 = \boxed{16}$.
 - b. The destination position of INFBN and transposition to IBFNN is at a shift of -5 from the intermediate position of 0(=26), $26 - 5 = \boxed{21}$.
 - c. The destination position of NSKGS and transposition to NGKSS is at a shift of -5 from the intermediate position of 5, $5 - 5 = \boxed{0}$.

See the chart of all 24 K4 shifts on the following page.

K4 (Plus Z) Caesar Shifts

0
ZOBKR
UOXOGHULBSOLIFBBWFLRVQQPRNGKSSO
TWTQSJQSSEKZZWATJKLUDIAWINFBNYP
VTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR

0
ZOBKR
UOXOGHULBSOLIFBBWFLRVQQPRNGKSSO
TWTQSJQSSEKZZWATJKLUDIAWINFBNYP
VTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR

1
APCLS
VPYPHIVMCTPMJGCCXGMSWRRQSOHLTP
UXURTKRTTFLAAXBUKLMVEJBXJOGCOZQ
WUUNAGQLXHELAYUKDEJHLVIVBVFDBS

2
BQDMT
WQZQIJWNUQNKHDDYHNTXSSRTPIMUUQ
VYVSULSUUGMBBYCVLMNWFKCYKPHDPAR
XVVOBHRMYIFMBZVLEFKIMWJWCWGMECT

3
CRENU
XRARJKXOEVROLIEEZIOUYTTSUQJNVVR
WZWTVMTVVHNCCZDWMNOXGLDZLQIEQBS
YWWPCISNZJGNCAWMFGLJNXKXDHNFDU

4
DSFOV
YSBSKLYPFWSMPJFFAJPVZUUTVRKOWWS
XAXUWNWUIODDAEXNOPYHMEAMRJFRCT
ZXXQDJTOAKHODBXNGHMKOYLIEYIOGEV

5
ETGPW
ZTCTLMZQGXTQNGGBKQWAVVUWSLPXXT
YBYVXOVXXJPEEBFYOPQZINFENSKGSDU
AAYYREKUPBLIPECYOHINLPZMFZJPHFW

6
FUHQX
AUDUMNARHYUROLHHCLRXBWWVXTMQYYU
ZCZWPWYYKQFFCGZPQRAJOGCOTLHTEV
BZZSFLVQCMJQFDZPIJOMQANAGAKQIGX

7
GVIRY
BVEVNOBSIZVSPMIIDMSYCCXWYUNRZZV
ADAXZQXZZLRGGDHAQRSBKPHDPUMIUFW
CAATGMWRDNKRGEAQJKNRBOHBBLRJHY

8
HWJSZ
CWFOWPCTJAWTQNJJENTZDYYXZVOSAAB
BEBYARYAAMSHHEIBRSTCLQIEQVNVJVGX
DBBUHNSXSEOLSHFBRKLQOSPCICMSKIZ

9
IXKTA
DXGXPQDUKBXUROKKFOUAEZZYAWPTBBX
CFCZBSZBBNTIIFJCTUDMRJFRWOKWHY
ECCVIOYTFPMTIGCSLMRPTDQJDNTLJA

10
JYLUB
EYHYQREVLCYVSPLLGPVBFAAZBXQUCCY
DGDACTACCOUJJGKDTUVEENSKGSXPLXIZ
FDDWJJPZUGQNUJHDTMNSQUEREKEOUMKB

11
KZMVC
FZIZRSFWMZWTQMMHQWCGBBACYRVDDZ
EHEBDUBDDPVKKHLEUVWFOTLHTYQMYJA
GEEKQAVHROVKIEUNOTRVFSFLFPVNL

12
LANWD
GAJASTGXNEAXURNNIRXDHCCBDZSWEEA
FIFCEVCEEQWLLIMFVWXGPMIUIZRZNB
HFFYLRLBWISPWLFJFVOPUSWGTGMGQWOMD

13
MBOXE
HBKBTUHYOFBYVSOOJSYEIDDCBATXFFB
GJGDFWDFFRKMMJNGWXYHQVNJVASOALC
IGGZMSCXJTXMKMGWPQVTXHUHNHRXPNE

14
NCPYF
ICLCUVIZPGCZWTPPKTZFEEDFBUYGGC
HKHEGXECCSYNNKOHXYZIRWOKWBTBMD
JHHANTDYKURYNLHXQRWUYIVIOISYQOF

15
ODQZG
JDMDVWJAQHDAXUQQLUAGKFFEGCVZHHD
ILIFHYFHHTZOLPIYZAJSPXLXUQCNE
KIIBOUEZLVZOMIYRSXVZJWJPJTZRPG

16
PERAH
KENEWXKBRIEBYVRMVBHLGGFHDWAIIE
JMJGIZGIIUAPPMQJZABKTYQMYDVRDOF
LJJCPVFAMWTAJNZSTYWAKXKQKUSQH

17
QFSBI
LFOFXYLCSJFCZWSSNWCIMHHGIEBXJFJ
KNKHJAHJJVBQQNRKABCLUZRNEWSEPG
MKKDQWGBNXUBQOKATUZXBLYLRLVBTRI

18
RGTCJ
MGPYZMDTKGDAXTTOXDJNIIHJFYCKKG
LOLIKBIKKWCRROSLBCDMVASOAFXTFQH
NLLERXHCYOYVCRPLBUVAYCMZMSMWCUSJ

19
SHUDK
NHQHZANEULHEBYUUPYEKOJJIKGZDLLH
MPMJLCJLLXDSSPTMCDENWBTPBGYUGRI
OMMFSYIDPZWDQMCVWBZDNANTNXDVTK

20
TIVEL
OIRIABOFVMIFCZVVQZFLPKKJLHAEMMI
NQNKMDKMMYETTQUNDEFQXCUQCHZVHSJ
PNNGTZJEQAXETRNWDXCAEOBOUYEWUL

21
UJWFM
PJSJBCPGWNJGDAWWRAGMQLLKMIBFNNJ
OROLNELNNZFURVOEFGPYDVRDIAWITK
QOOHUAKFRBYFUSOEYDBFPCVPZFXVM

22
VBXGN
QKTKCDQHXOKHEBXXSBHNRMLNJCBOOK
PSPMOFMOOAGVSWPFGHQZEWSEJBXJUL
RPPIVBLGSCZGVTFFYZECGQDQWQAGYWN

23
WLYHO
RLULDERIYPLIFCYTTCIOSNNMOKDHPPL
QTQNPNGNPPBHWWTXQGHIRAFXTFKCYKVM
SQQJWCMTDAHWUQGZAFDHRERXRBHZXO

24
XNZIP
SMVMEFSJZQMJDZJZUDJPTOONPLEIQQM
RUROHQOQCIXUYRHIJSBGYUGLDZLWN
TRRXDNIEBIXVRHABGEISFSYSCAIYP

25
YNAJQ
TNWNFGTKARNKHEAAVEKQUPPOQMFJRRN
SVSPRIPRRDJYYVZSIJKTCHZVHMEAMXO
USSLYEOJVFCJYWSIBCHFJTGTZTDBZQ

QOOHUAKFR Note: Beginning Q answers question at end of K3 CT. OOH is missing part of Hoover's name as one travels around the corners of the ciphertext on the sculpture. The Q + UAKFR approximates the word QUAKER.

REKUPBLIPECYOHIN approximates the word REPUBLICAN.

Herbert Hoover was a QUAKER and a REPUBLICAN. Not J. EDGAR, but Herbert is Sanborn's first name.

The above is evidence that the forgoing procedures were done correctly.

Important: Subsequent analysis has demonstrated that, in the zero-shift version, DIAWI lies "fully exposed", INFBN is "subtly shaded" and NGKSS, the "nuance of illusion", lies between subtle shading and nothingness (the absence of light). The implication here is to locate the version of this string that lies in the fully exposed position, the position that DIAWI occupies in the zero-shift version. This location is found in the shift of 10 which begins with JYLUB.